# PS4 - Question 6

# Does the rank of Phd program matter?

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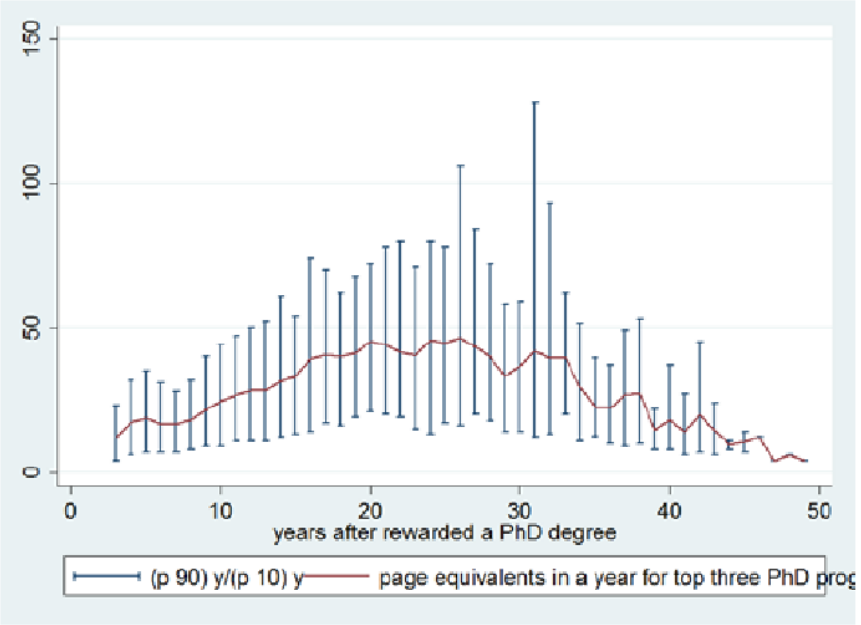


**Figure 1.2 Average productivity of phuzicists from top PhD programs**

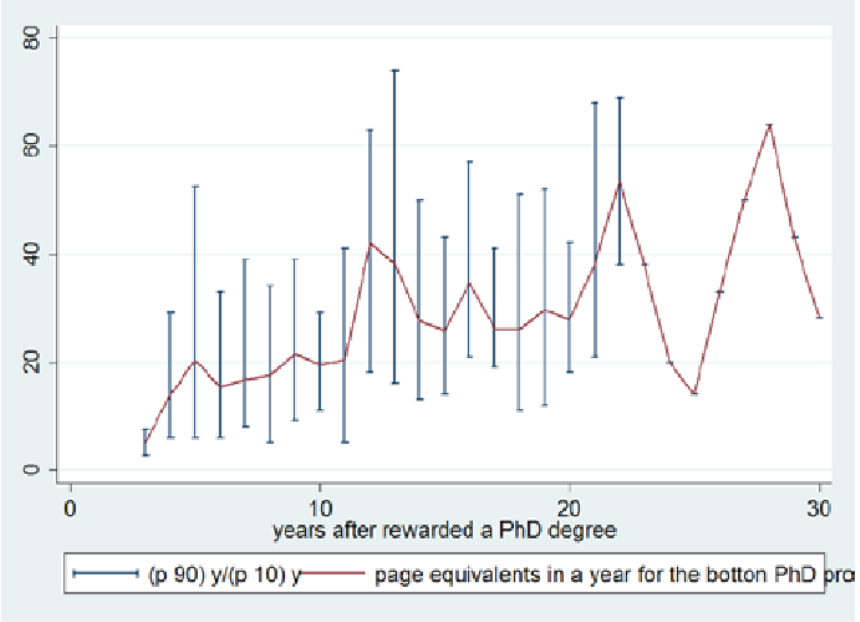


**Figure 1.3 Average productivity of phuzicists from bottom PhD programs**

From the original model: $logy_{it} = \alpha_i + \rho_{1i} log(y_{i,t-1}) + \rho_{2i} log(y_{i,t-2}) + \beta_1 e_{it} + \beta_2 e_{it}^2 + \beta_3 \frac{1}{e_{it} * rank_i} + \beta_4 sex_i + \beta_5 d80_i + u_{it}$

HTIV give the estimate of $\beta_3$ as 2.169, with standard error 0.439, which is significant at 1% level.

Then we may calculate the effect of Phd program rank as:

$\frac{\widehat{dy}}{drank} = -\frac{\widehat{\beta}_3 \bar{y}}{\bar{e}_{it} * \overline{rank_i}^2} = 0.582$, with confidence interval (0.55, 0.61)

Change the model a little bit:

$$logy_{it} = \alpha_i + \rho_{1i} log(y_{i,t-1}) + \rho_{2i} log(y_{i,t-2}) + \beta_1 e_{it} + \beta_2 e_{it}^2 + \beta_4 sex_i + \beta_5 d80_i + \gamma_1 rphd_i + u_{it}$$

And

$$logy_{it} = \alpha_i + \rho_{1i} log(y_{i,t-1}) + \rho_{2i} log(y_{i,t-2}) + \beta_1 e_{it} + \beta_2 e_{it}^2 + \beta_4 sex_i + \beta_5 d80_i + \gamma_1 rphd_i + \gamma_2 rphd_i * e_{it} + u_{it}$$

Run the HTIV estimate

# Results for the first model with rphd as dependent variable

| y | Coef. | Std. Err. | z | P>\|z\| | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| TVexogenous | | | | | | |
| exp | .1334863 | .0039177 | 34.07 | 0.000 | .1258077 | .1411649 |
| expsq | -.0027482 | .0001023 | -26.87 | 0.000 | -.0029486 | -.0025477 |
| TVendogenous | | | | | | |
| y1 | .5503531 | .0123463 | 44.58 | 0.000 | .5261548 | .5745514 |
| y2 | -.4245096 | .011859 | -35.80 | 0.000 | -.4477529 | -.4012664 |
| TIexogenous | | | | | | |
| d80 | .0200051 | .0703131 | 0.28 | 0.776 | -.117806 | .1578162 |
| sex | -.0038279 | .0664573 | -0.06 | 0.954 | -.1340818 | .126426 |
| TIendogenous | | | | | | |
| rphd | .0106467 | .1643817 | 0.06 | 0.948 | -.3115355 | .3328289 |
| _cons | 1.493955 | .5041824 | 2.96 | 0.003 | .5057757 | 2.482134 |
| sigma_u | .27206981 | | | | | |
| sigma_e | .45018136 | | | | | |
| rho | .26753137 | (fraction of variance due to u_i) | | | | |

Note: TV refers to time varying; TI refers to time invariant.

## Results for the second model with rphd and rage = rphd*exp as dependent variable

| y | Coef. | Std. Err. | z | P>\|z\| | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| TVexogenous | | | | | | |
| exp | .1336193 | .0039771 | 33.60 | 0.000 | .1258242 | .1414143 |
| expsq | -.0027481 | .0001023 | -26.87 | 0.000 | -.0029485 | -.0025476 |
| TVendogenous | | | | | | |
| y1 | .5503426 | .0123475 | 44.57 | 0.000 | .5261418 | .5745433 |
| y2 | -.4244933 | .0118604 | -35.79 | 0.000 | -.4477393 | -.4012474 |
| rage | -.0000455 | .0002339 | -0.19 | 0.846 | -.000504 | .000413 |
| TIexogenous | | | | | | |
| d80 | .0198638 | .0702159 | 0.28 | 0.777 | -.1177568 | .1574844 |
| sex | -.0039444 | .066376 | -0.06 | 0.953 | -.1340389 | .12615 |
| TIendogenous | | | | | | |
| rphd | .010819 | .1641611 | 0.07 | 0.947 | -.3109308 | .3325688 |
| _cons | 1.493525 | .5034936 | 2.97 | 0.003 | .5066953 | 2.480354 |
| sigma_u | .2715974 | | | | | |
| sigma_e | .4501797 | | | | | |
| rho | .26685227 | (fraction of variance due to u_i) | | | | |

Note: TV refers to time varying; TI refers to time invariant.